CisternLab · Impact Evidence Brief

ESTIMATED RESULT · ARCHIVED HISTORICAL REANALYSIS — ERA5

Edirne, Edirne, Republic of Türkiye · 2023–2025 · three independent complete calendar years

Scenario & decision

Mode: archive. Roof 120 m² · efficiency 80% · non-potable demand 250 L/day · current tank 2,000 L. Starting storage: **0 L independently in each period**. Years are never concatenated.

Selected alternative: 3,000 L. Recommended tested capacity: 3,000 L. Target: 90% of each period's maximum standard tested supply.

Modeled resource benefit

> Potential freshwater displacement equals modeled harvested rainwater delivered ONLY IF each liter replaces an otherwise used liter of freshwater. This is a conditional estimate, not measured savings, municipal consumption reduction, carbon reduction or a financial benefit.

Unused tank storage is not already displaced freshwater. Remaining freshwater requirement means modeled unmet demand, assuming it is met by another freshwater source.

2023 · 365 days

Identical rain, roof, demand and zero starting water; volumes in L

Case / tank L	Delivered L	Coverage %	Unmet L	Overflow L	Unused end L
No harvesting / 0	0	0	91,250	44,188.8	0
Current / 2,000	39,030.8	42.77	52,219.2	5,158	0
Recommended tested / 3,000	40,995.6	44.93	50,254.4	3,193.2	0
Selected alternative / 3,000	40,995.6	44.93	50,254.4	3,193.2	0

Selected potential freshwater displacement: **40,995.6 L (41 m³)**, conditional. Full-demand days: 141; shortage days: 224; longest shortage streak: 57 days.

Selected minus current delivered: 1,964.8 L; reduction in unmet demand: 1,964.8 L; coverage change: 2.15 percentage points. Recommended minus current delivered: 1,964.8 L; reduction in unmet demand: 1,964.8 L; coverage change: 2.15 percentage points.

Selected minus no-harvesting modeled tank overflow: -40,995.6 L. This is not verified drainage runoff or flood-risk reduction.

2024 · 366 days

Identical rain, roof, demand and zero starting water; volumes in L

Case / tank L	Delivered L	Coverage %	Unmet L	Overflow L	Unused end L
No harvesting / 0	0	0	91,500	49,872	0
Current / 2,000	41,299.2	45.14	50,200.8	7,928.8	644
Recommended tested / 3,000	43,547.6	47.59	47,952.4	4,680.4	1,644
Selected alternative / 3,000	43,547.6	47.59	47,952.4	4,680.4	1,644

Selected potential freshwater displacement: **43,547.6 L (43.55 m³)**, conditional. Full-demand days: 153; shortage days: 213; longest shortage streak: 29 days.

Selected minus current delivered: 2,248.4 L; reduction in unmet demand: 2,248.4 L; coverage change: 2.46 percentage points. Recommended minus current delivered: 2,248.4 L; reduction in unmet demand: 2,248.4 L; coverage change: 2.46 percentage points.

Selected minus no-harvesting modeled tank overflow: -45,191.6 L. This is not verified drainage runoff or flood-risk reduction.

2025 · 365 days

Identical rain, roof, demand and zero starting water; volumes in L

Case / tank L	Delivered L	Coverage %	Unmet L	Overflow L	Unused end L
No harvesting / 0	0	0	91,250	40,080	0
Current / 2,000	34,261.6	37.55	56,988.4	5,792.4	26
Recommended tested / 3,000	36,144	39.61	55,106	3,910	26
Selected alternative / 3,000	36,144	39.61	55,106	3,910	26

Selected potential freshwater displacement: **36,144 L (36.14 m³)**, conditional. Full-demand days: 123; shortage days: 242; longest shortage streak: 51 days.

Selected minus current delivered: 1,882.4 L; reduction in unmet demand: 1,882.4 L; coverage change: 2.06 percentage points. Recommended minus current delivered: 1,882.4 L; reduction in unmet demand: 1,882.4 L; coverage change: 2.06 percentage points.

Selected minus no-harvesting modeled tank overflow: -36,170 L. This is not verified drainage runoff or flood-risk reduction.

Three-year SUM of independent annual selected-tank estimates: 120,687.2 L (120.69 m³) delivered; total demand 274,000 L; remaining demand 153,312.8 L. Longest within-year shortage streak: 57 days. Annual differences reflect rainfall timing as well as totals; this is historical sensitivity, not a continuous multiyear tank or a climate trend.

Methodology & limitations

Smallest positive tested standard capacity reaching the target fraction of maximum standard tested supply; for three years it must satisfy every annual target. Tolerance: 1e-10 × each period's maximum supply. No relative recommendation when any period has zero achievable supply.

Coverage is used / demand; undefined with zero demand. Shortage means any positive daily unmet volume; streaks reset at annual boundaries and are not meteorological droughts. Signed coverage differences are percentage points. Display values are rounded; source data and engine calculations retain full precision.

Daily rain-before-demand model; fixed demand and efficiency; independent empty starts. No sub-daily precision, drainage routing, treatment, first flush, evaporation or leakage. Actual implementation depends on installation, water quality, actual usage and relevant regulations. Three years cannot establish long-term climate reliability or trends.

Evidence & reproduction

Reproduce with the recorded original daily rainfall, physical inputs, selected capacity, target and empty starts, using the unchanged simulation, sizing and variability engines. Retain the original JSON or existing CSV/JSON downloads alongside this brief. See README.md and BUILD_05_REPORT.md for the pure-function reproduction procedure.

ARCHIVED HISTORICAL REANALYSIS — ERA5 · Edirne, Edirne, Republic of Türkiye · 2023–2025 · three independent complete calendar years

Preserved during BUILD 04 · retrieval date: 23 October 2026 · no new live request. Original file: docs/verification/build-04-live-20261023T192935527Z-snapshot.json. Browser delivery is a byte-for-byte copy; original calculated result tables are ignored.

2023 · 365 complete daily records

Source	Open-Meteo · ERA5 · rain_sum · mm · validated
Coverage	2023-01-01 to 2023-12-31
Location identity	Edirne, Edirne, Republic of Türkiye · 2023 · ID 747712
Requested coordinates	41.67719, 26.55597
Returned ERA5 grid	41.75, 26.5
Daily aggregation	Europe/Istanbul / GMT+3, fixed UTC offset 10800 s
Original retrieval	2026-10-23T19:29:37.494Z
Source request	https://archive-api.open-meteo.com/v1/archive?latitude=41.67719&longitude=26.55597&start_date=2023-01-01&end_date=2023-12-31&daily=rain_sum&models=era5&precipitation_unit=mm&timezone=auto&timeformat=iso8601

2024 · 366 complete daily records

Source	Open-Meteo · ERA5 · rain_sum · mm · validated
Coverage	2024-01-01 to 2024-12-31
Location identity	Edirne, Edirne, Republic of Türkiye · 2024 · ID 747712
Requested coordinates	41.67719, 26.55597
Returned ERA5 grid	41.75, 26.5
Daily aggregation	Europe/Istanbul / GMT+3, fixed UTC offset 10800 s
Original retrieval	2026-10-23T19:29:37.558Z
Source request	https://archive-api.open-meteo.com/v1/archive?latitude=41.67719&longitude=26.55597&start_date=2024-01-01&end_date=2024-12-31&daily=rain_sum&models=era5&precipitation_unit=mm&timezone=auto&timeformat=iso8601

2025 · 365 complete daily records

Source	Open-Meteo · ERA5 · rain_sum · mm · validated
Coverage	2025-01-01 to 2025-12-31
Location identity	Edirne, Edirne, Republic of Türkiye · 2025 · ID 747712
Requested coordinates	41.67719, 26.55597
Returned ERA5 grid	41.75, 26.5
Daily aggregation	Europe/Istanbul / GMT+3, fixed UTC offset 10800 s
Original retrieval	2026-10-23T19:29:37.299Z
Source request	https://archive-api.open-meteo.com/v1/archive?latitude=41.67719&longitude=26.55597&start_date=2025-01-01&end_date=2025-12-31&daily=rain_sum&models=era5&precipitation_unit=mm&timezone=auto&timeformat=iso8601

Original and browser-copy SHA-256: `2aee94b5a369acf8164bbb1e9d7265c8b89f5eba0a2a8719fe4a132c5e7197d7`

Weather data by Open-Meteo · ERA5 / Copernicus C3S · CC BY 4.0 · Locations: GeoNames. Original rain is preserved; CisternLab transforms it into modeled estimates. No provider endorsement or station-measured rainfall is claimed.

General background: EPA Environmental Benefits of Green Infrastructure; EPA Prepare for Drought; Open-Meteo Historical Weather API and licence. These sources support harvesting/modeling context, not independent verification of CisternLab outputs.

https://www.epa.gov/green-infrastructure/environmental-benefits-green-infrastructure
https://www.epa.gov/green-infrastructure/prepare-drought
https://open-meteo.com/en/docs/historical-weather-api
https://open-meteo.com/en/licence